University of Amsterdam

Recommender Systems in Society

WasteLess

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Students:
Liam Boesten
15819965

Oliver Martin
15827488

Sebastian Rociu
15617085

Diaz Wiersma
15379078

Tutor:
Simon Pauw and Dina Strikovic

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1 Introduction

Food waste is a serious problem, not only environmentally but also economically and socially. Worldwide, households waste an estimated 79 kilograms of food per person each year. This points to a broader problem in the way modern food systems are organized and used (United Nations Environment Programme 2024). The Netherlands fits within this wider picture. In 2022, Dutch consumers wasted around 33.4 kilograms of solid food per person per year. So when food is thrown away, it is not just the food itself that is lost, but also the raw materials, labour, and energy that were needed to produce it (Voedingscentrum 2024). The effects of food waste also reach far beyond the kitchen. Food loss and waste are estimated to account for 8–10% of global greenhouse gas emissions and add unnecessary pressure on land, water, and biodiversity (Food and Agriculture Organization of the United Nations n.d. United Nations Environment Programme 2024). Reducing food waste is therefore not a minor issue, but an important sustainability challenge.

By turning everyday grocery behaviour into machine-readable data, this system participates in the wider datafication of human life, where daily routines become a continual source of data (Mejias and Couldry 2019; Strikovic 2026a).

This report will be about a content-based recipe recommender system that was developed to help reduce food waste at household level. The basic idea is straightforward. Instead of recommending recipes in general, the system recommends recipes based on what a user already has at home and on the rating of the recipe. To do that, it combines pantry data and shelf-life information with recipes from Albert Heijn Allerhande, an online platform with a wide selection of everyday meals, themed dishes, and ingredient-based recipe ideas (Albert Heijn n.d.). In practice, users create a digital pantry using supermarket receipt data, which allows ingredients, quantities, and expiration dates to be tracked. The system then ranks recipes by looking at how well they match the pantry, while also giving extra weight to ingredients that are close to expiring. In that sense, the goal is not only to make meal planning easier. It is also to encourage more sustainable food use by prioritizing recipes that may help prevent avoidable waste (Food and Agriculture Organization of the United Nations n.d. United Nations Environment Programme 2024).

The main research question in this report is: **To what extent can a content-based recommendation system based on pantry data and shelf-life information contribute to reducing food waste, and which ethical and theoretical considerations play a role in**

this process? To answer that question, five sub-questions are discussed. The first focuses on how ingredient names from recipes can be normalized and linked reliably to a user's pantry. The second looks at the combination of pantry coverage and urgency weighting, and the trade-offs involved in prioritizing ingredients that are close to expiring. The third examines how diversity in recipe recommendations can be maintained, and how this relates to the tension between accuracy and diversity. The fourth addresses possible forms of bias in both the dataset and the algorithm, and what these mean for the fairness of the system. The final sub-question considers which ethical principles matter in the design of a personal food recommender system and how these can be safeguarded in practice.

2 Method

2.1 Data Gathering

To collect recipe data, a web scraper was developed in Python targeting the Albert Heijn Allerhande recipe platform. The scraper operates in three phases: it authenticates the allerhande API using an anonymous access token, retrieves a list of recipe URLs by paginating through the search endpoint, and finally downloads and parses each recipe page using BeautifulSoup. For each recipe, the scraper extracts the name, description, preparation steps, required kitchen tools and original URL. The results are exported to a structured Excel file. A total of 2000 recipes were collected, which was deemed sufficient given the time constraints of the project (two weeks). A short delay was introduced between requests to avoid overloading the server. Albert Heijn Allerhande was chosen as the data source because it is one of the most widely used Dutch recipe platforms and offers a broad and representative selection of everyday recipes in the Netherlands. Alternative datasets were considered but were found to be predominantly US-oriented, with measurements and product references that do not transfer to a Dutch supermarket context. Figure 1 shows the first three rows of the resulting spreadsheet with recipes.

1	ID	Naam	Beschrijving	Ingrediënten	Stappen	Benodigdheden	
2	1202387	Bloemisalade met whipped geitenkaas	De bloemi in deze gezonde, glutenvrije salade met whipped geitenkaas wordt licht geroosterd, terwijl de romige geitenkaascrème met citroen, munt en peterselie zorgt voor frisheid. Lekker met knapperig amandelschaafsel en in 20 minuten op tafel.	50 g radijsjes 30 ml appelcideraazijn 50 ml kokend water 320 g bloemi 1 el milde olijfolie 2 el amandelschaafsel 1 tl venkelzaad 15 g verse platte peterselie 10 g verse munt 150 g pure geitenkaas 45+ 200 g zuivelspread naturel 1 biologische citroen	1. Snijd de radijs in dunne plakken. Doe in een kom met de appelcideraazijn en het kokende water en laat staan tot gebruik. 2. Verhit ondertussen de grillpan zonder olie of boter op hoog vuur. Snijd de bloemi in de lengte tot gelijke roosjes en meng met de olie. Gril in delen rondom 5-6 min. Verhit een koekenpan zonder olie of boter op middelhoog vuur en rooster het amandelschaafsel 3-4 min. Laat afkoelen op een bord. Rooster in dezelfde pan het venkelzaad 2 min. 3. Pluk ondertussen de blaadjes van de peterselie en snijd de takjes fijn. Pluk ook de muntblaadjes. Doe de takjes van de peterselie, ½ van de blaadjes van de peterselie en ½ van de muntblaadjes, de geitenkaas en zuivelspread in de hoge beker van de staafmixer en pureer glad. Boen de citroen schoon, rasp de gele schil en pers de helft van de vrucht uit. Meng het citroenrasp en 2 el -sap (per 4 personen) door de whipped geitenkaascrème. Breng op smaak met peper en eventueel zout. 4. Smeer de whipped geitenkaascrème uit over een groot bord en verdeel de bloemi, radijs, amandel, het venkelzaad en de rest van de kruiden erover.	grillpan staafmixer met hoge beker	https://www.ah.nl/allerhande/recept/
3	1202381	Eiersalade met tuinerwten	Vervang de mayonaise in je eiersalade 'ns door cottagecheese voor een luchtig en fluweelzacht resultaat. Samen met de knapperige tuinerwten en citroenrasp is deze eiersalade de perfecte combi voor wie houdt van een frisse spread.	200 g diepvries tuinerwten 12 middelgrote scharreleieren 15 g verse bieslook 20 g verse platte peterselie 20 g verse dille 1 biologische citroen 100 g cottagecheese	1. Laat de tuinerwten ontdooien. Kook de eieren in 10 min. hard. Blancheer de tuinerwten 5 min. in kokend water. Giet af en spoel koud af. Snijd ondertussen de verse kruiden fijn. Boen de citroen schoon, rasp de gele schil en pers een halve citroen uit. 2. Prak de eieren en tuinerwten fijn met een vork. Meng met de cottagecheese, verse kruiden, het citroenrasp, 1 el -sap (per 12 stuks), versgemalen zwarte peper en eventueel zout.		https://www.ah.nl/allerhande/recept/
	1202372	Gezond paaselbroodje met spinazie	Dit gezonde paaselbroodje met spinazie is een verrassend hartig paasrecept. In een luchtig volkorendeeg verstop je romige ricotta, geskonken spinazie en een zachtgekookt ei. Perfect voor een feestelijke paasbrunch, maar ook heerlijk als voedzame lunch.	3.5 g gedroogde gist 125 ml lauwwarm water 5 middelgrote scharreleieren 250 g volkoren tarwemeel 2 ¾ el milde olijfolie 150 g prei ½ tl gedroogde Italiaanse kruiden 400 g spinazie 1 volkoren beschuit 125 g ricotta 50 g geraspte Goudse belegen 30+ kaas ¼ tl gemalen nootmuskaat 2 el maanzaad 2 el wit sesamzaad	1. Los de gist op in 50 ml lauwwarm water en laat 10 min. staan. Klop 1 ei los. Doe het meel in de kom van de staande mixer. Voeg het gistmengsel, 1 el olie, de helft van het losgeklopte ei en de rest van het lauwwarme water toe. Meng 10 min. met de mixer tot een soepel deeg dat niet plakt. 2. Haal het deeg uit de kom. Verdeel in 4 gelijke porties en bol de porties op door de randen naar onderen te vouwen, het deeg een kwartslag te draaien, en de handeling te herhalen. Vet ze rondom in met een ½ el olie, verdeel over een met bakpapier beklede bakplaat en dek af met een schone, vochtige theedoek. Laat 1 uur rijzen op kamertemperatuur. 3. Kook ondertussen de rest van de eieren in 6 min. halfzacht en laat schrikken onder koud stromend water. Snijd de prei zo fijn mogelijk. Verhit de rest van de olie in een hapjespan en fruit de prei 5 min. Voeg de Italiaanse kruiden en spinazie in delen toe en laat slinken. Laat het preispinaziemengsel uitlekken in een vergiet, druk het vocht eruit en laat afkoelen. Verkruimel het beschuit en meng met de ricotta, kaas en nootmuskaat door het groentemengsel. Breng op smaak met peper en eventueel zout. 4. Verwarm de oven voor op 200 °C. Pel de eieren. Rol een bol deeg uit tot een cirkel van Ø 20 cm. Verdeel ¼ van het spinaziemengsel en 1 ei erover, laat 3 cm van de randen vrij. Vouw de spinazie om het ei dicht en vouw het deeg over de spinazie tot een eivorm. Verwijder overtollig deeg en druk de naden dicht. Maak zo nog 3 broodjes. Verdeel de broodjes met de naad naar beneden over	staande mixer met deeghaken bakpapier	https://www.ah.nl/allerhande/recept/

Figure 1: The first 3 rows of the recipe spreadsheet

Because collecting real pantry data from actual users within the two-week project timeframe was not doable, synthetic pantry data was generated for 100 simulated users. The proxy algorithm draws from the ingredient list extracted from the recipe dataset and organises ingredients into categories such as dairy, fresh vegetables, and dry goods. Each ingredient is assigned a unit of measurement, a realistic quantity range, and a probability of being present in a pantry, ensuring that common staples appear (like sugar and egg) more frequently than niche items (like kimchi). To introduce diversity, five household profiles were defined: families, students, couples, single-person households and health-conscious eaters. Each profile carries a different likelihood

factor, reflecting realistic differences in shopping behaviour, budget and household size. Quantities are sampled in appropriate and realistic steps. Grams are in multiples of 50 and loose items are whole numbers. The resulting dataset was saved as an Excel file and used with the recommender algorithm. Although synthetic data avoids some immediate privacy risks, it still encodes our own design assumptions about what a "typical" Dutch household looks like, which means that data bias can already be introduced at the data-generation stage (Strikovic 2026b). Figure 2 shows the first two rows of the user pantries.

user_i	profiel	categorie	ingredient	hoeveelhei	eenhei
1	stel	zuivel	ei	6	stuk
1	stel	zuivel	scharrelei	4	stuk
1	stel	zuivel	melk	1350	ml
1	stel	zuivel	halfvolle melk	700	ml
1	stel	zuivel	volle melk	350	ml
1	stel	zuivel	boter	200	g
1	stel	zuivel	roomboter	200	g
1	stel	zuivel	Griekse yoghurt	500	g
1	stel	zuivel	crème fraîche	150	g
1	stel	zuivel	geraspte kaas	100	g
1	stel	zuivel	mozzarella	150	g
1	stel	zuivel	feta	200	g
1	stel	zuivel	Griekse feta	100	g
1	stel	zuivel	roomkaas	125	g
1	stel	groente_vers	ui	4	stuk
1	stel	groente_vers	rode ui	2	stuk
1	stel	groente_vers	knoflook	3	stuk
1	stel	groente_vers	paprika	1	stuk
1	stel	groente_vers	wortel	2	stuk
1	stel	groente_vers	aardappel	300	g
1	stel	groente_vers	kastanjechampignon	200	g
1	stel	groente_vers	sla	2	stuk
1	stel	groente_vers	rucola	125	g
1	stel	groente_vers	prei	1	stuk
1	stel	groente_vers	bloemkool	1	stuk
1	stel	groente_vers	citroen	4	stuk
1	stel	groente_vers	limoen	3	stuk
1	stel	groente_vers	cherrytomaat	250	g
1	stel	groente_vers	venkel	1	stuk
1	stel	groente_vers	asperge	500	g
1	stel	groente_vers	ijsbergsla	1	stuk
1	stel	groente_vers	komkommer	2	stuk

Figure 2: An example of 2 users from the pantry proxy

2.2 Ingredient Matching

The first step is linking recipe ingredients to a user's personal pantry. Because ingredient names in natural language vary considerably, all text is normalised before comparison. Stop words that carry no semantic content are removed, including origin descriptors (e.g. "Italian") and preparation styles (e.g. "smoked"). The remaining tokens are stemmed using a Dutch Snowball stemmer (Porter 1980), so that morphological variants such as *chicken fillets* and *chicken* are mapped to the same root form.

After normalisation, all unique ingredient names from the recipe database are resolved against the user's pantry via a hash map in which normalised ingredient strings act as keys and associated quantities as values. A hash map stores each pantry item under its normalised name, so that looking up an ingredient reduces to computing that same normalised form and checking whether the resulting string exists as a key. This is an operation that runs in constant time regardless of pantry size. Because both recipe ingredients and pantry items undergo the same stemming procedure, an exact key lookup suffices: the normalised form of a recipe ingredient either matches a pantry key or it does not, with no ambiguity. This approach helps avoid the false positives that probabilistic distance-based methods can produce, such as a match between *chicken* and *chickpeas* (Cohen, Ravikumar, and Fienberg 2003). Quantities are converted to a common base unit (grams, millilitres, or piece equivalents) to enable comparison with required recipe amounts where necessary.

A recipe is eligible for recommendation only if at least one of its ingredients is fully covered by the pantry, or at least two ingredients are partially covered, preventing recipes from ranking highly on urgency alone when the user holds almost none of the required items.

In the implementation, this is labeled as 'almost done' recipes: recipes for which a large portion of the required ingredients are already present in the user's pantry. This label makes it immediately clear which recipes can be prepared with minimal additional shopping, improving both convenience and waste reduction.

2.3 Scoring Formula

The scoring pipeline consists of three consecutive stages: a base score, a length-adjusted score, and a final waste-boosted score.

2.3.1 Pantry Coverage

For each ingredient i in a recipe of N ingredients, the per-ingredient coverage is defined as:

$$\text{coverage}_i = \min\left(1.8,\ u(d_i) \cdot \frac{q_i^{\text{avail}}}{q_i^{\text{req}}}\right) \tag{1}$$

where q_i^{avail} is the available quantity, q_i^{req} is the required quantity, and $u(d_i)$ is the urgency multiplier defined in the next section. The cap at 1.8 prevents a single highly urgent ingredient from distorting the recipe-level mean. The pantry coverage score for the recipe is:

$$\text{pantry_coverage} = \frac{1}{N}\sum_{i=1}^{N} \text{coverage}_i \tag{2}$$

Recipes with high pantry coverage are naturally more likely to be marked as 'almost done', reflecting their readiness for immediate preparation.

2.3.2 Base Score

The base score combines pantry coverage with a content-based similarity term. The user profile is represented as a TF-IDF vector constructed from the normalised pantry items and their associated category labels, enabling direct comparison with recipe ingredient vectors via cosine similarity (Salton and Buckley 1988):

$$\text{base} = 0.75 \times \text{pantry_coverage} + 0.25 \times \text{tfidf_similarity} \tag{3}$$

These weights reflect the assumption that pantry coverage is a stronger relevance signal than stylistic affinity. The TF-IDF term gives a small boost to recipes whose ingredients broadly match what the user tends to stock, even when quantities do not align exactly.

To validate the chosen weights, five configurations were evaluated on a subset of 20 users (Table 1). Coverage weight was varied from 0.00 to 1.00 in steps of 0.25, with TF-IDF set to the complement. The 0.75/0.25 split was selected based on the resulting almost-ready rate and average score at rank 1.

Coverage	TF-IDF	Avg. coverage top-3	Almost-ready (%)	Avg. score rank 1
1.00	0.00	0.374	73.3	0.381
0.75	**0.25**	**0.321**	**66.7**	**0.329**
0.50	0.50	0.255	73.3	0.264
0.25	0.75	0.171	73.3	0.180
0.00	1.00	0.072	20.0	0.088

Table 1: Sensitivity analysis over five weight configurations (n=20 users). Bold row indicates the configuration used in the final system.

2.3.3 Length Penalty and Waste Bonus

The length penalty discounts recipes with few ingredients, which would otherwise score disproportionately high simply by virtue of having few ingredients to miss:

$$\text{length_penalty} = \begin{cases} 1.0 & \text{if } n_{\text{ingredients}} \geq k \text{ or urgent_items} \geq 2 \\ 1 - 0.4 \cdot \dfrac{k - n_{\text{ingredients}}}{k} & \text{otherwise} \end{cases} \tag{4}$$

where $k = 6$ is the minimum ingredient threshold below which the penalty activates. The penalty is waived when a recipe covers at least two urgent ingredients, ensuring that short dishes making efficient use of near-expiry items are not unfairly suppressed.

The waste bonus scales the adjusted score proportionally to the fraction of urgent ingredients in the recipe, and is bounded between 1 and 1.35:

$$\text{waste_bonus} = 1 + 0.35 \times \frac{\text{urgent_items}}{N} \quad \in [1,\, 1.35] \tag{5}$$

These stages form a consecutive scoring pipeline, yielding the final score:

$$\text{final} = \text{base} \times \text{length_penalty} \times \text{waste_bonus} \tag{6}$$

After this an alternative ranking is made that blends the final score with the recipe's average user rating, with a maximum weight increase of 20%, providing users the option to balance waste reduction against recipe quality. Coverage scores are calculated for all recipes at once using NumPy arrays, making the process 10–30 times faster than computing them one by one in a Python loop (Van der Walt, Colbert, and Varoquaux 2011).

2.4 Urgency Weighting

Urgency weighting steers the scoring formula based on the remaining shelf life of pantry items. Each ingredient is assigned a multiplier $u(d)$ that increases as the expiry date approaches:

$$u(d) = \begin{cases} 2.2 & \text{if } d \leq 0 \\ 2.0 & \text{if } d = 1 \\ 1.8 & \text{if } d = 2 \\ 1.6 & \text{if } d \leq 4 \\ 1.4 & \text{if } d \leq 7 \\ 1.15 & \text{if } d \leq 14 \\ 1.05 & \text{if } d \leq 30 \\ 1.0 & \text{otherwise} \end{cases} \tag{7}$$

where d is the number of days remaining until expiry. The thresholds were chosen to produce a steep increase in urgency within the final days before expiry, loosely inspired by the idea of loss aversion from prospect theory: wasting food can be seen as a salient "loss" that feels worse the closer the deadline comes (Kahneman and Tversky 1979). Urgency values are calculated once for each pantry item and then reused for all recipes, reducing computation time.

2.5 Diversity

A well-documented shortcoming of score-driven recommender systems is that the result list can become dominated by closely related items, reducing the amount of choices for the user (Carbonell and Goldstein 1998). The system addresses this through two mechanisms.

First, Maximal Marginal Relevance (MMR) is used to compose the final ranking. MMR iteratively selects the next recipe as a weighted combination of its recommendation score and its maximum cosine similarity in the TF-IDF space to already selected recipes:

$$\text{MMR} = \lambda \cdot \text{score_final} - (1 - \lambda) \cdot \max_{r_j \in S} \text{sim}(r_i, r_j) \tag{8}$$

where S is the set of already selected recipes, $\text{sim}(r_i, r_j)$ denotes cosine similarity in the shared TF-IDF space, and $\lambda = 0.7$ expresses a moderate preference for relevance over diversity. Keeping λ as a free parameter allows the system to be tuned to the desired trade-off between accuracy and variety (Ziegler et al. 2005).

Second, a category cap is applied: at most three recipes per meal category may appear in the final list. Categories are assigned through rule-based keyword matching on recipe titles. This measure prevents the list from being dominated by a single meal type, which can give users the choice of recommendations across a full day. Recipes exceeding the category limit are moved to a reserve list and reinstated if space remains in the overall output.

2.6 Ethical and Sustainability Considerations

The system is designed with sustainability as its primary rationale. By prioritising recipes that use ingredients already present in the pantry, and by boosting ingredients close to expiry, the recommender directly targets avoidable household food waste. This aligns with the ethical principle of beneficence: generating outcomes that extend beyond individual convenience to broader environmental benefit (Wright 2011). At the same time, reducing waste has a justice dimension, as lower-income households suffer proportionally greater financial consequences from discarded food (European Commission 2024). Given that food waste contributes significantly to global emissions, this design also reflects sustainability goals highlighted in recent policy and research reports (FAO 2021; United Nations Environment Programme 2024).

Several risks were identified and addressed in the design. First, the Albert Heijn Allerhande dataset introduces a clear form of data bias: the recipe set predominantly reflects Western Dutch cuisine, underrepresenting dishes common among Turkish, Surinamese, Indonesian, and Moroccan communities (Strikovic 2026b). This can lead to unequal representation and reduced relevance for certain user groups. In addition, system bias may emerge through ranking mechanisms such as MMR, where certain cuisines or popular recipes receive more exposure, reinforcing existing imbalances. While the re-ranking and category cap partially mitigate this by promoting

diversity, full mitigation would require dataset augmentation. Second, the datafication of grocery behaviour raises questions of control and power, as translating everyday actions into data inherently shapes what can be measured and optimized (Mejias and Couldry 2019). To address privacy concerns, pantry data is stored locally by default, ensuring that individual consumption patterns are not externally shared. Third, the scoring weights and urgency thresholds represent human design choices, which may in edge cases produce unintuitive or suboptimal recommendations. To support user autonomy and transparency, each recommendation includes an explanation of its ranking, such as the number of expiring ingredients it incorporates.

3 Results

Based on the export of 100 users and 1000 recommendations (aanbevelingen.xlsx in GitHub), the recommender shows a clear pattern: the highest-ranked positions contain the recipes that best match the available pantry stock and are therefore the most immediately usable. As shown in Figure 3, the average final score of the number 1 recommendation is 0.628, gradually declining to 0.408 at position 10. This indicates that the ranking is not random, but that the recipes at the top of the list systematically score better than the rest of the top 10.

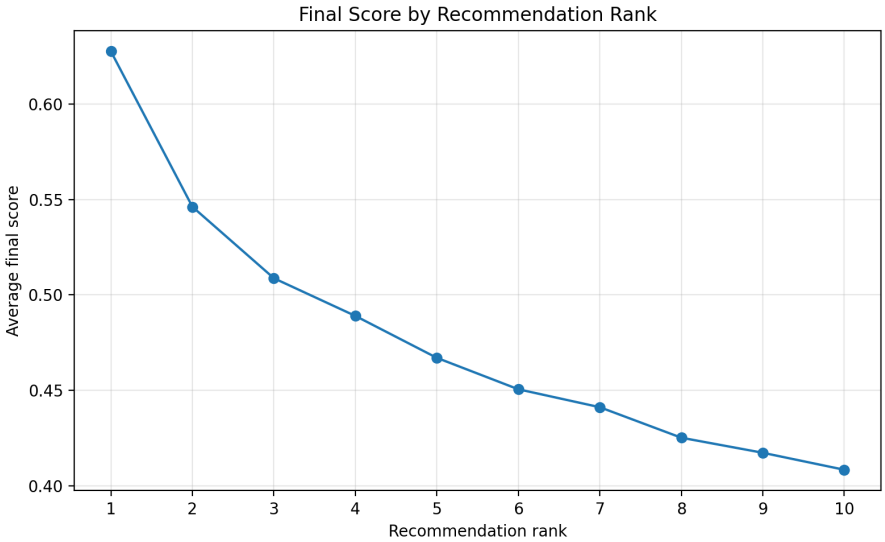

Figure 3: Final Score by Recommendation Rank. The average final score decreases with each rank, showing that the ordering consistently distinguishes between stronger and weaker recommendations.

The top-ranked recommendations also perform best in terms of ingredient match. As shown in Figure 4, within the top 3, the average coverage is 49.7%, compared with 43.4% across the full top 10. In addition, 76.0% of the top 3 recommendations are marked as *almost ready*, compared with 54.0% across the full list. This suggests that the recommender gives priority to recipes for which users already have a large share of the required ingredients at home. By doing this, the system demonstrates that it prioritizes recipes that are closer to being fully preparable from the current pantry, improving convenience and reducing potential waste.

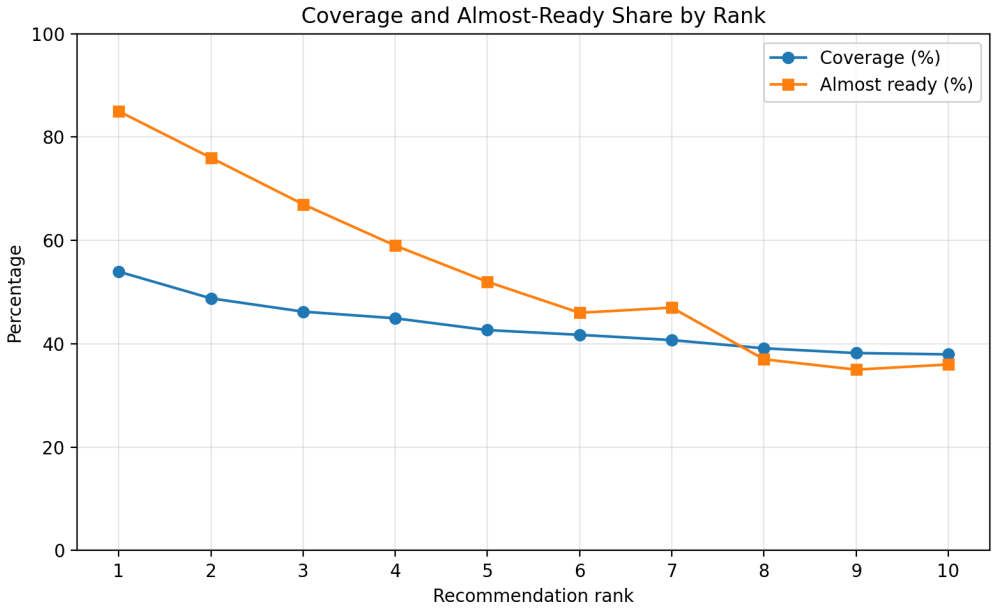

Figure 4: Coverage and Almost-Ready Share by Rank. The highest-ranked recommendations use more ingredients that are already available and more often consist of recipes that can be prepared almost immediately.

When recipe rating is included in the scoring, the average score increases across all recommendation positions, as shown in Figure 5. On average, the score rises from 0.478 to 0.545, an increase of 0.067 points. The effect is strongest at the top of the ranking: the average score of the number 1 recommendation increases from 0.628 to 0.716 when rating is taken into account. This indicates that recipe rating does not fundamentally change the ranking logic, but it does provide an additional boost to recipes that are not only well matched to the pantry, but also better appreciated overall. In that sense, including recipe rating makes the final ranking slightly more quality-sensitive, while preserving the original focus on pantry overlap and usability.

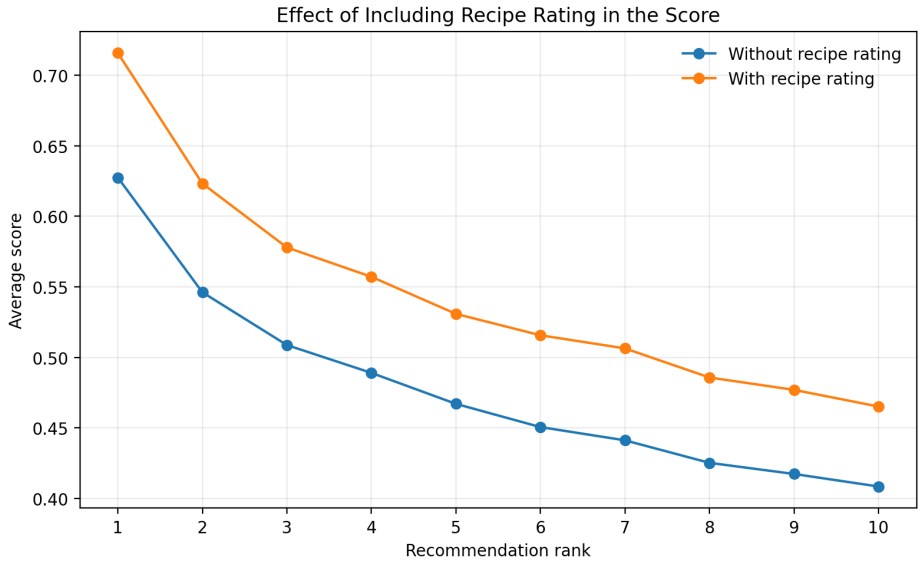

Figure 5: Effect of Including Recipe Rating in the Score. Across all recommendation positions, the average score is higher when recipe rating is included, with the largest increase visible at the top of the ranking.

At the same time, the top 10 still allows room for variation. On average, a recommended recipe contains 7.0 relevant pantry matches, of which 2.2 are urgent. The average waste bonus is greater than 1 (mean 1.115), showing that recipes containing products that need to be used sooner do in fact receive an extra boost in the ranking.

The results do vary somewhat by user profile. As shown in Figure 6, *family* has the highest average final score (0.541), while *student* has the lowest (0.404). This suggests that the recommender performs best for profiles with a broader or better-matching pantry, whereas profiles with less overlap naturally receive lower scores.

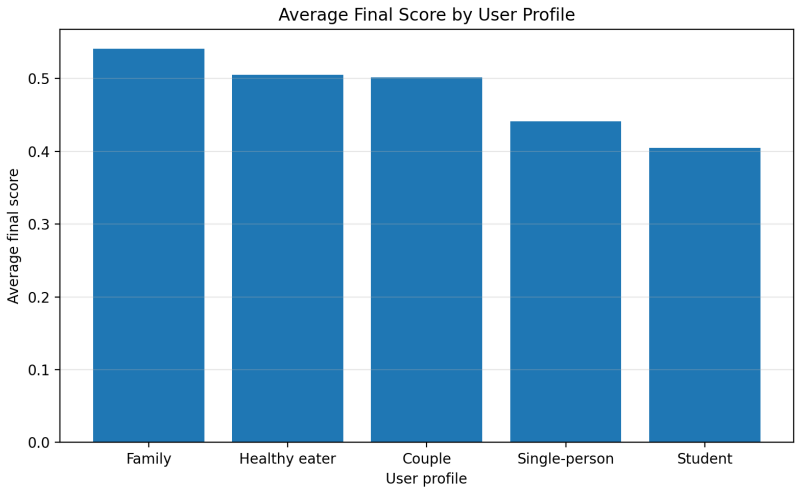

Figure 6: Average Final Score by User Profile. There are differences between user profiles, but across all profiles the recommender follows the same logic: recipes with greater overlap and higher usability rank higher.

4 Conclusion

This study shows that a content-based recommender system using pantry data and shelf-life information can meaningfully contribute to reducing household food waste. The results demonstrate that prioritizing pantry coverage over stylistic similarity, combined with urgency weighting, produces recommendations that are both actionable and waste-reducing. Ingredient matching through stemming and hash map lookup proves reliable in practice, while MMR reranking and category caps successfully maintain diversity without sacrificing relevance (Carbonell and Goldstein 1998; Ziegler et al. 2005).

The system is not without limitations. The Allerhande dataset reflects a Western Dutch food context, which introduces data bias and reduces relevance for users with different culinary backgrounds (Strikovic 2026b). This bias, combined with the use of synthetic pantry data, means the results may not fully generalize to real-world use. From an ethical perspective, the system steers user behaviour through its scoring design, which raises questions of transparency and autonomy that are only partially addressed by the current explanation mechanism (Mejias and Couldry 2019; Wright 2011).

Overall, the approach is promising, but its societal value depends on whether it can be validated with real receipt data and a more culturally diverse recipe corpus. Future work should also examine whether the urgency thresholds and scoring weights transfer across different household contexts and user groups.

5 Discussion

The results confirm that the scoring pipeline translates its design goal into measurable outcomes: higher pantry coverage and almost-ready rates at the top of the ranking show that the system consistently prioritizes actionable recipes. However, the performance gap between household profiles suggests that optimizing for average performance may obscure meaningful inequalities. A system that works well for families but less well for students does not reduce food waste equally across user groups, which connects directly to the justice dimension raised in the ethics section.

The sensitivity analysis shows that coverage dominance is the critical design decision. When TF-IDF is used exclusively, the almost-ready rate collapses to 20%, confirming that content-based similarity alone is insufficient for real-world usability. This points to a broader insight for sustainability-oriented recommender systems: feasibility should be treated as a first-class objective, not a secondary signal.

Future work should validate the system with real receipt data and a more culturally diverse recipe corpus, as both the synthetic pantry proxies and the Allerhande dataset encode assumptions that may not transfer to all user groups.

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